Exploration, UCB, and Action-Value Learning

1 Introduction

Reinforcement Learning (RL) learns from interaction. An agent takes actions, receives rewards, and tries to improve its decisions over time. The main goal is: maximize total reward.

One of the simplest and most important models in RL is **k-armed bandit problem**. In this:

- There are k possible actions.
- Each action gives a reward drawn from an unknown distribution.
- The agent must learn which action gives the highest expected reward.

Although it looks simple, it introduces a difficulty known as the **exploration–exploitation dilemma**.

2 Exploration–Exploitation Dilemma

For each action a, the true value is defined as:

$$q^*(a) = \mathbb{E}[R_t \mid A_t = a]$$

This represents the expected reward of choosing action a. But, the true value $q^*(a)$ is unknown.

So we maintain an estimate:

$$Q_t(a)$$

At every time step, the agent must decide between:

- **Exploitation:**

$$A_t = \arg\max_a Q_t(a)$$

  Choose the action that currently looks best.

- **Exploration:** Choose a different action to gather more information.

If the agent always exploits, it may get stuck with a suboptimal action. If it explores too much, it wastes reward.

To measure performance, we define **regret**:

$$\text{Regret}(T) = \sum_{t=1}^{T} (q^*(a^*) - q^*(A_t))$$

where a^* is the optimal action.

A good algorithm keeps regret growing as slowly as possible.

3 Upper Confidence Bound (UCB)

The Upper Confidence Bound (UCB) algorithm provides a principled way to balance exploration and exploitation.

The action selection rule is:

$$A_t = \arg\max_a \left[Q_t(a) + c\sqrt{\frac{\ln t}{N_t(a)}} \right]$$

where:

- $Q_t(a)$ is the estimated value of action a,
- $N_t(a)$ is the number of times action a has been selected,
- t is the current time step,
- $c > 0$ controls exploration.

The formula has two parts:

- $Q_t(a)$ encourages exploitation.
- $c\sqrt{\frac{\ln t}{N_t(a)}}$ adds an exploration bonus.

If an action has been chosen only a few times, $N_t(a)$ is small, and the bonus becomes large. If an action has been chosen many times, the bonus becomes small.

This method is based on the idea of optimism under uncertainty. It assumes that actions with high uncertainty might be better than they currently appear.

UCB achieves logarithmic regret:

$$\text{Regret}(T) = O(\log T)$$

which means the performance improves steadily over time.

4 Optimistic Initialization

Optimistic initialization is a simple but effective exploration strategy.

Normally, we initialize action values as:

$$Q_1(a) = 0$$

In optimistic initialization, we set:

$$Q_1(a) = Q_0$$

where Q_0 is a large positive value.

At the beginning, all actions appear good. The agent tries different actions to verify their values. Poor actions quickly receive lower estimates after observing real rewards.

This naturally forces exploration without randomness.

However, this method works best when the environment is stationary. If reward distributions change over time, optimistic initialization may not adapt well.

5 Incremental Update Methods

To estimate action values, we use averages.

If action a has been selected n times with rewards $R_1, R_2, \ldots, R_n$, then:

$$Q_n(a) = \frac{1}{n} \sum_{i=1}^{n} R_i$$

Storing all rewards is inefficient. So we use an incremental update formula:

$$Q_{n+1} = Q_n + \frac{1}{n}(R_n - Q_n)$$

This update only needs:

- The previous estimate Q_n
- The new reward R_n

A more general update rule is:

$$Q_{n+1} = Q_n + \alpha(R_n - Q_n)$$

where α is the learning rate.
If:

$$\alpha = \frac{1}{n}$$

we get the sample average.
If α is constant:

$$Q_{n+1} = (1 - \alpha)Q_n + \alpha R_n$$

This gives more weight to recent rewards. Such updates are useful in non-stationary environments where rewards change over time.

6 Conclusion

The exploration–exploitation dilemma is one of the most important ideas in reinforcement learning. At every step, the agent must decide whether to stick with what currently seems best or try something new that might lead to better rewards in the future. Obtaining the right balance between these two is our goal.

Methods like UCB give a clear mathematical way to manage this balance by accounting for uncertainty.

Optimistic initialization encourages early exploration in a simple but clever manner, and incremental update rules make value estimation practical and better for memory efficiency.